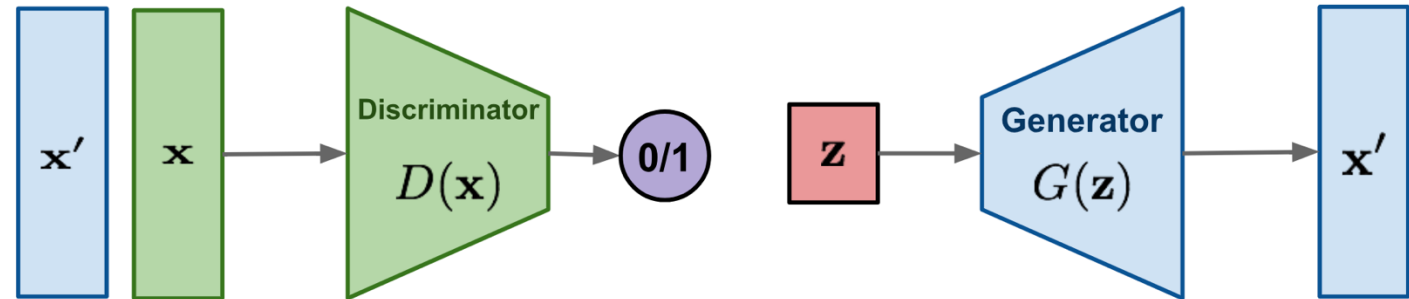


Диффузионные модели

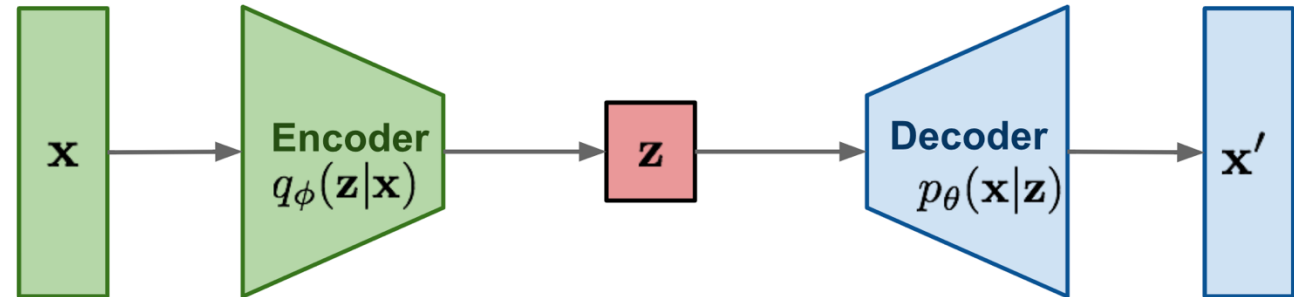


Что мы уже знаем

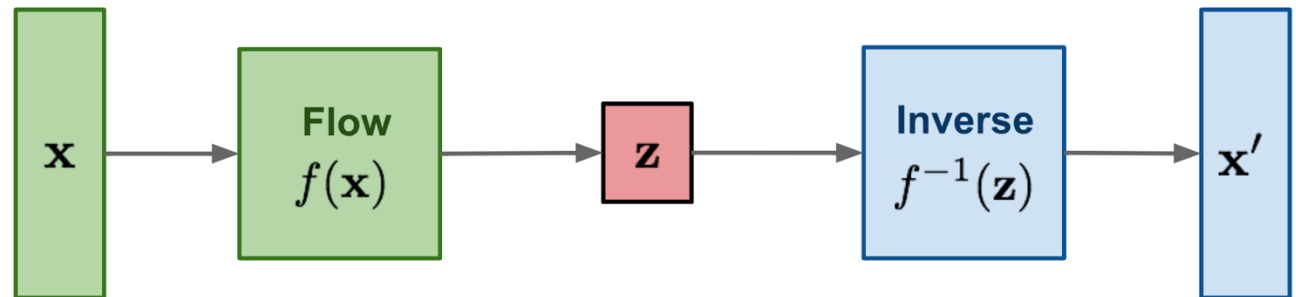
GAN: minimax the classification error loss.



VAE: maximize ELBO.



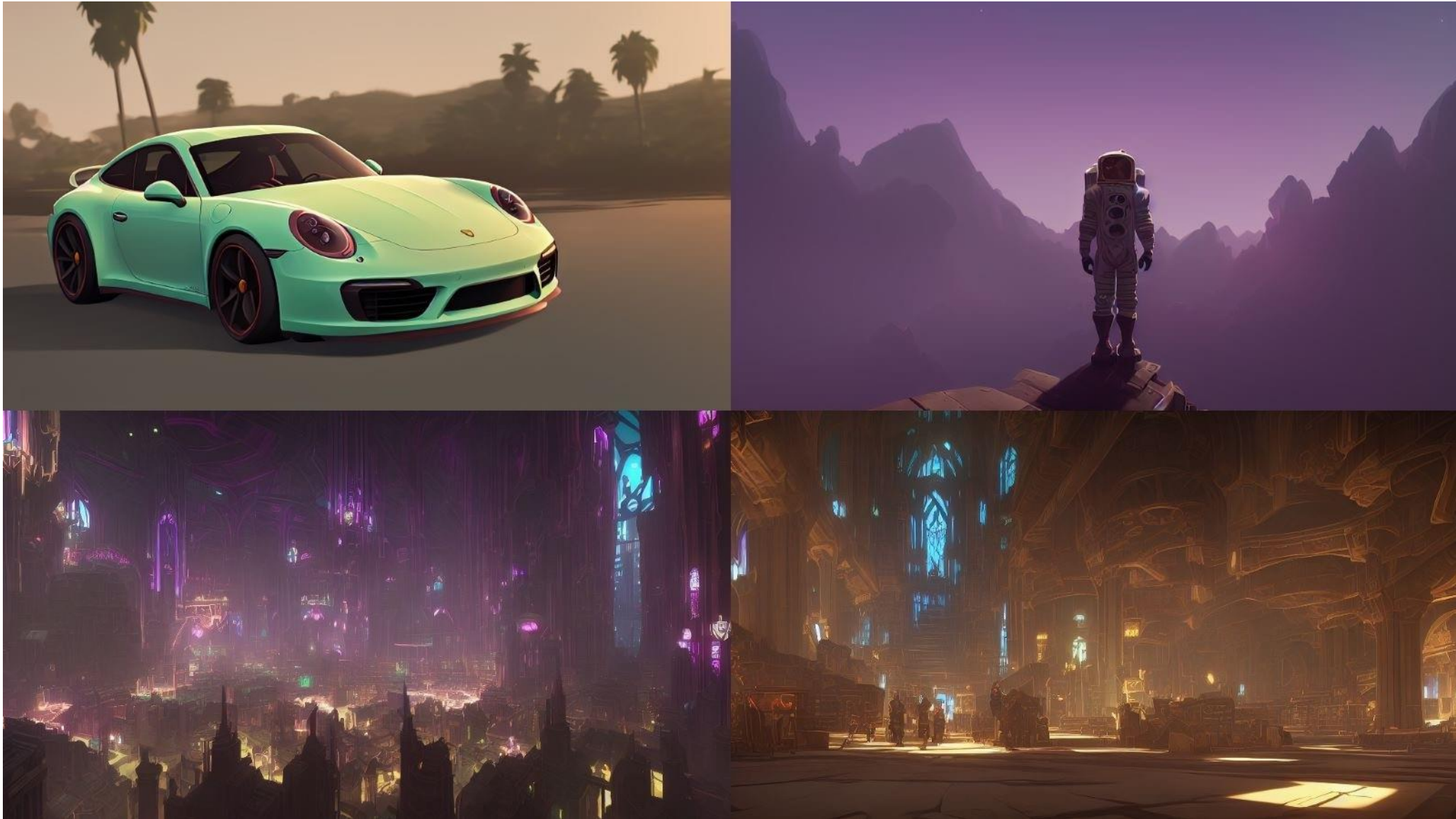
Flow-based generative models:
minimize the negative log-likelihood



Приложения



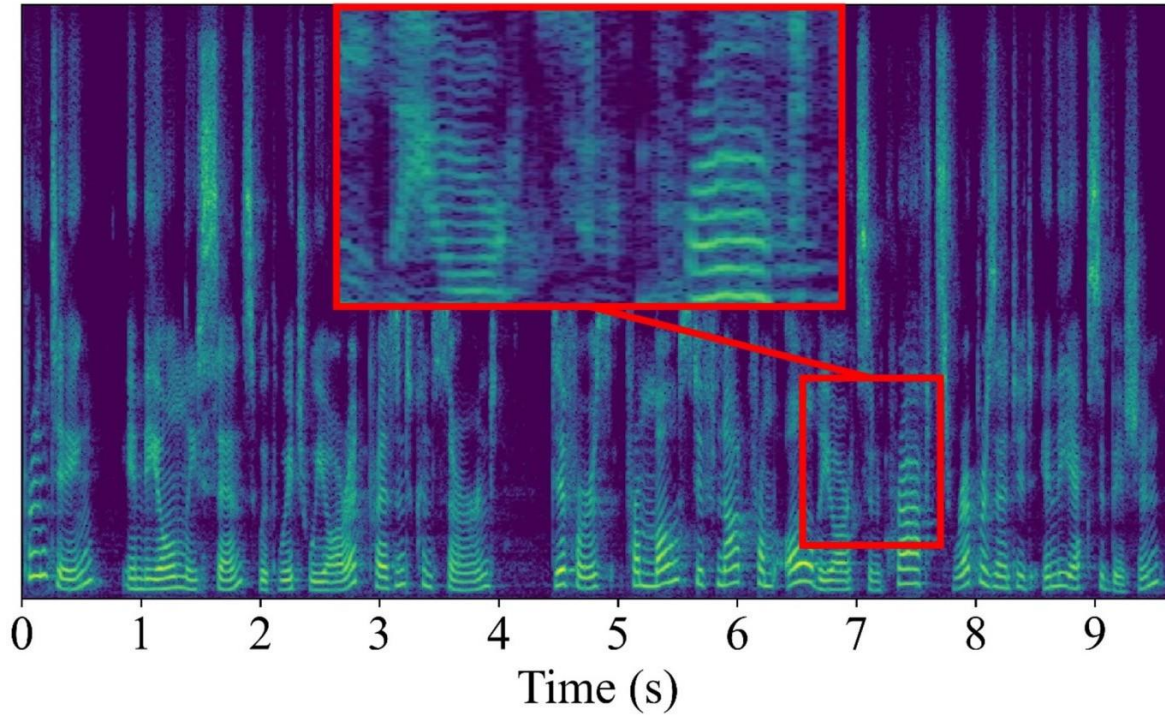
Stable Diffusion (StabilityAI)



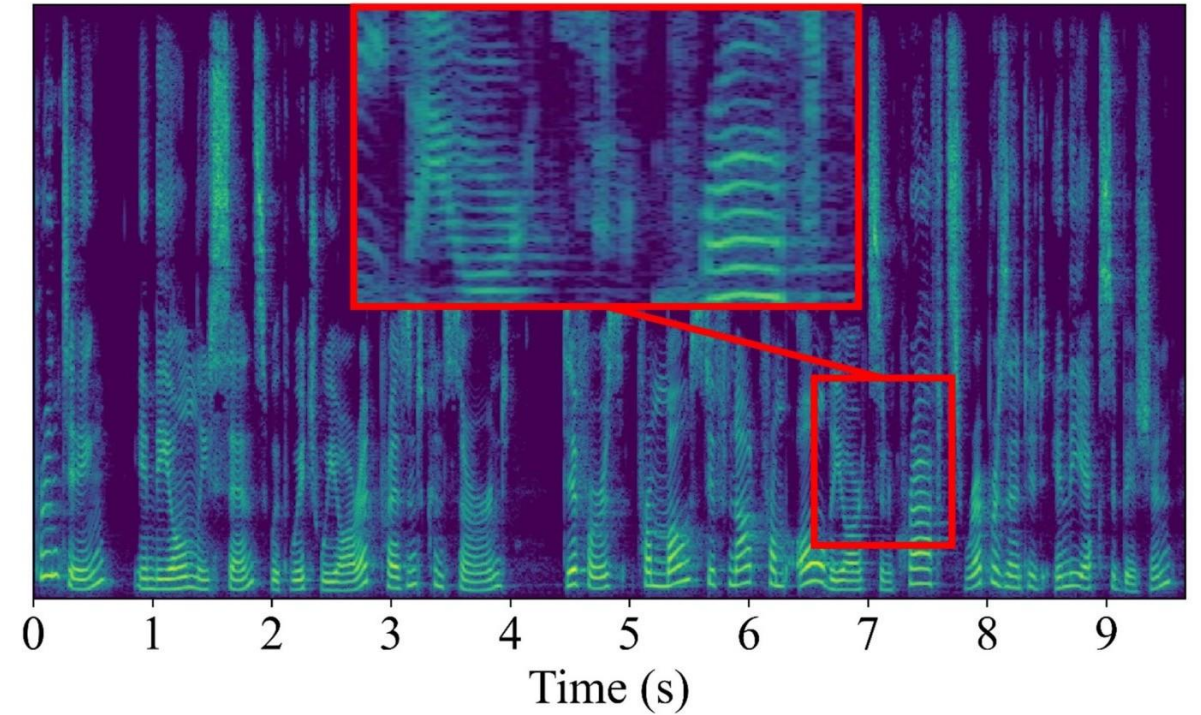
Source: <https://learnopencv.com/image-generation-using-diffusion-models/>

Text-to-speech generation

InferGrad

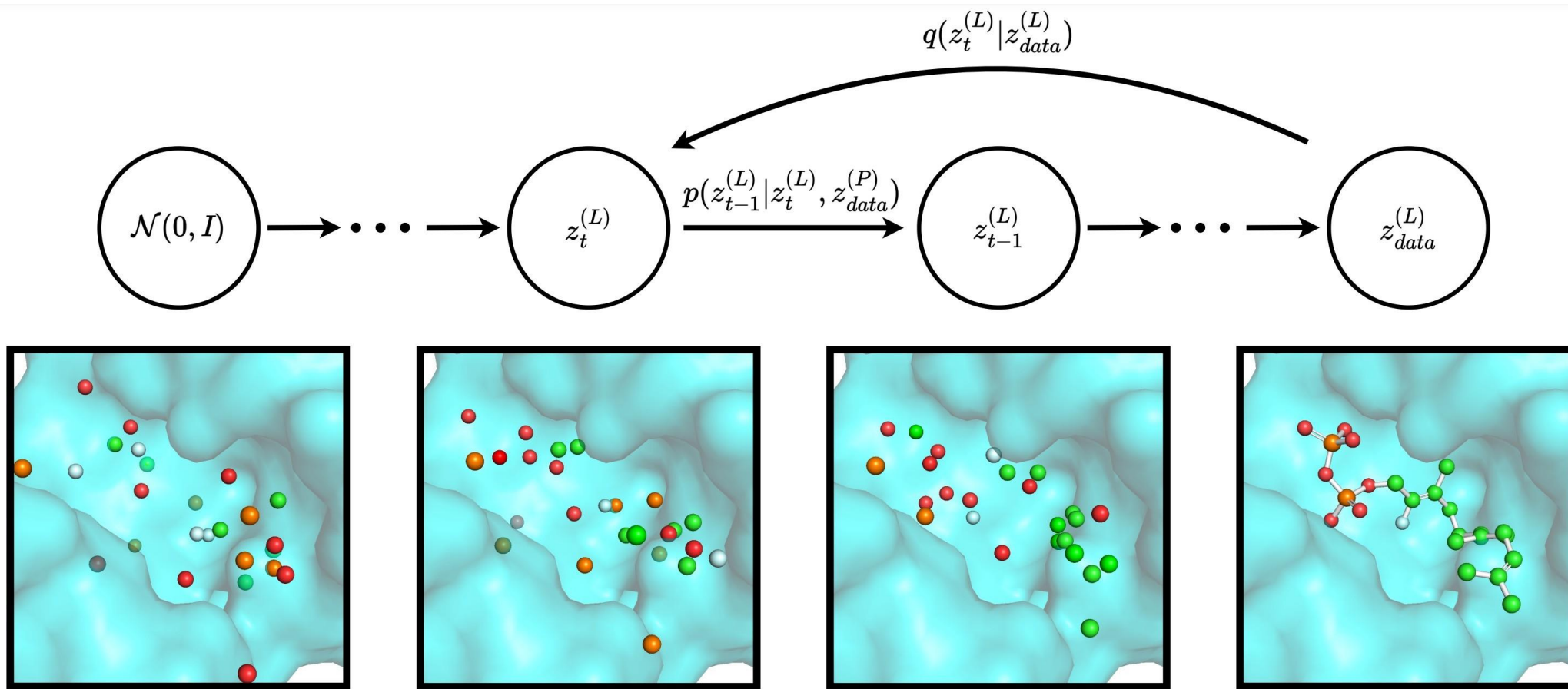


Ground Truth



Source: <https://github.com/heejkoo/Awesome-Diffusion-Models#text-to-speech>

Molecular and material generation



Source: <https://github.com/heejkoo/Awesome-Diffusion-Models#molecular-and-material-generation>

Astronomical spectra generation

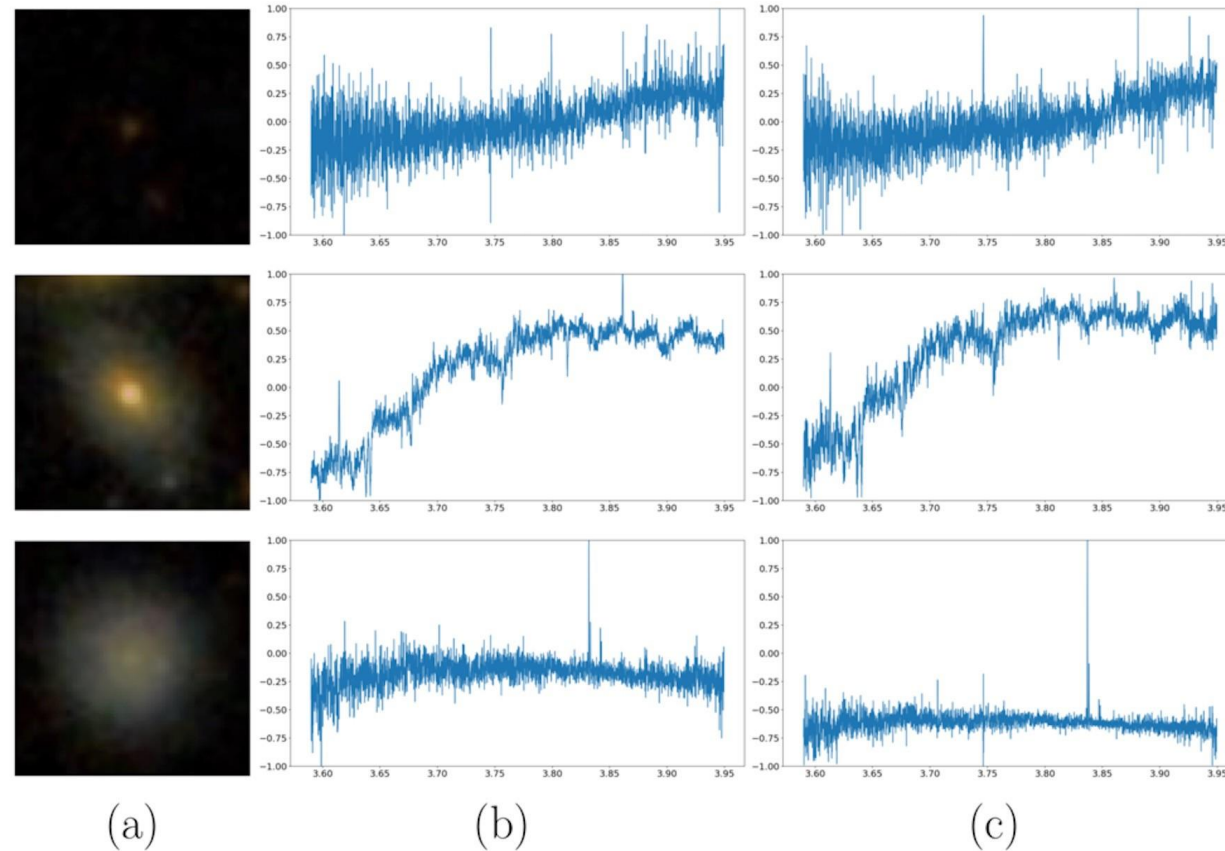


Figure 3: Generated spectra for the images in (a). In (b) we show the real spectra, in (c) the best match according to our contrastive model, out of 25 samples.

Source: https://ml4physicalsciences.github.io/2022/files/NeurIPS_ML4PS_2022_78.pdf

Dark matter density modelling

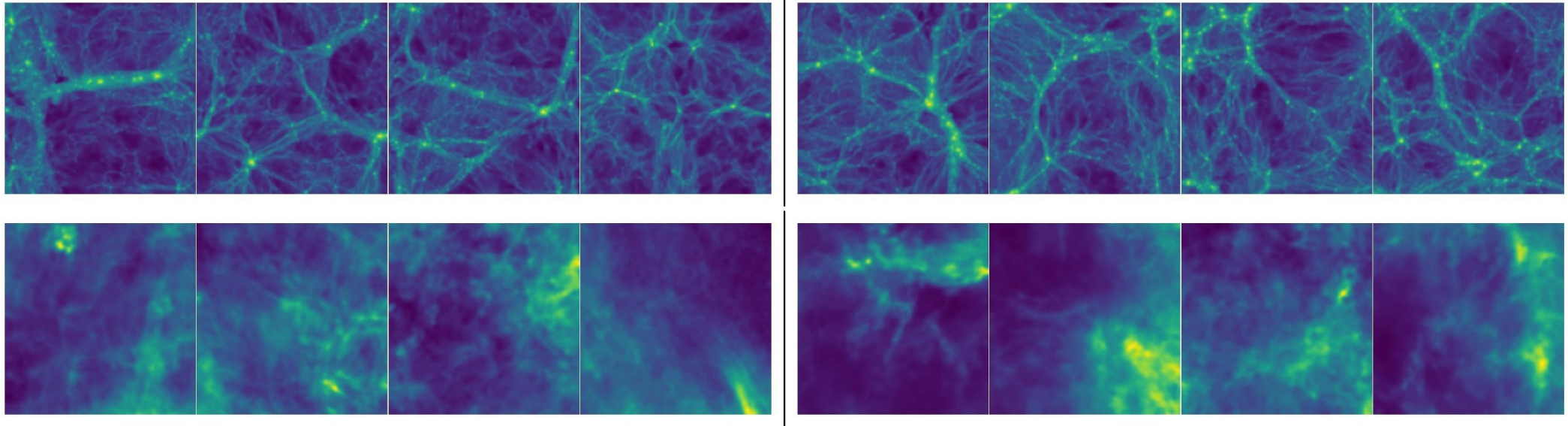


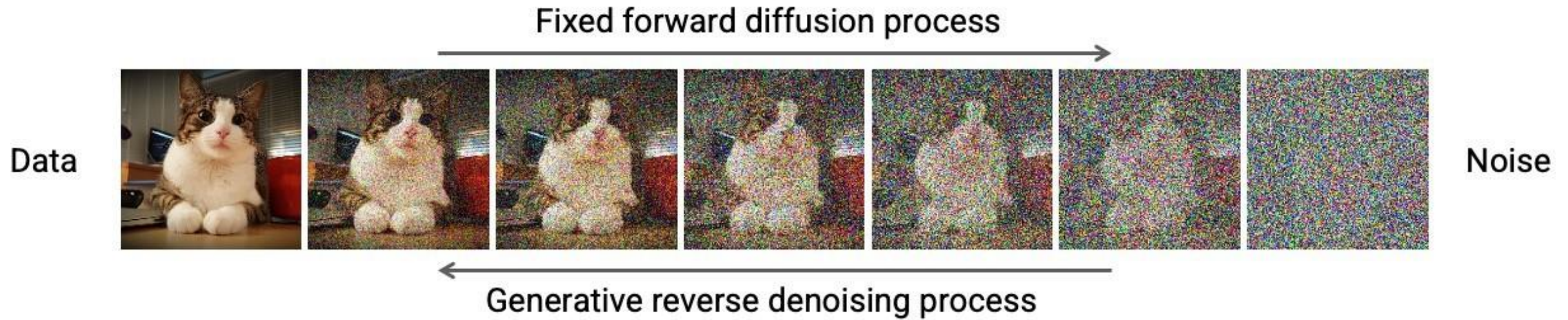
Figure 1: Four log cold dark matter mass density fields from the training data (top left) and from the sampled model (top right) at 128×128 . Four samples of dust from the training data (bottom left) and from the trained model (bottom right).

Source: https://ml4physicalsciences.github.io/2022/files/NeurIPS_ML4PS_2022_25.pdf

Интуиция

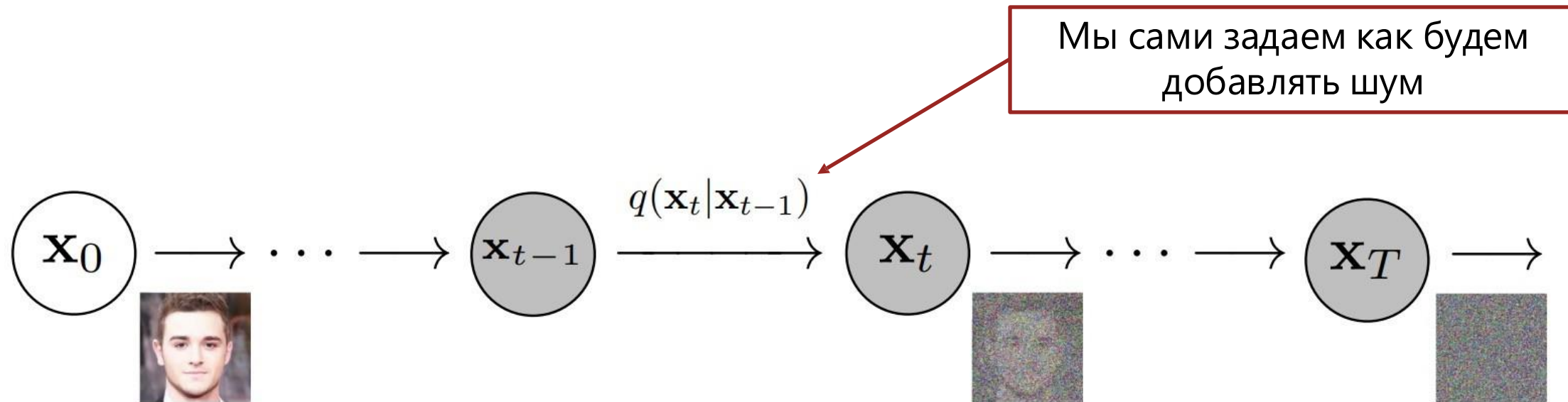


Общая идея диффузионных моделей



Источник: <https://cvpr2022-tutorial-diffusion-models.github.io/>

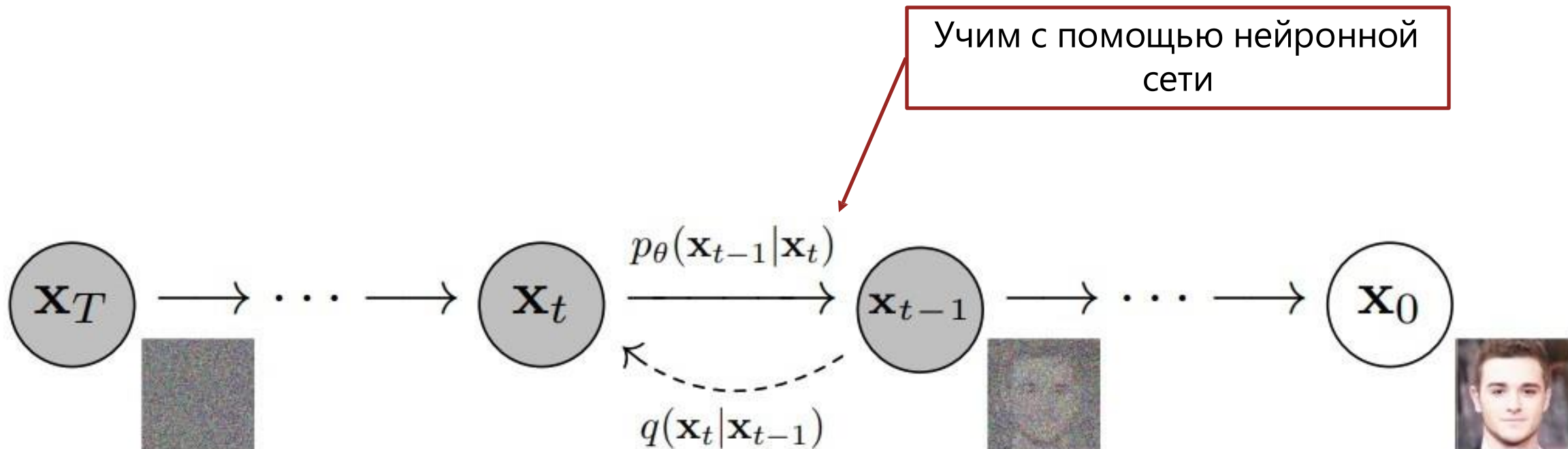
Процесс диффузии (зашумления)



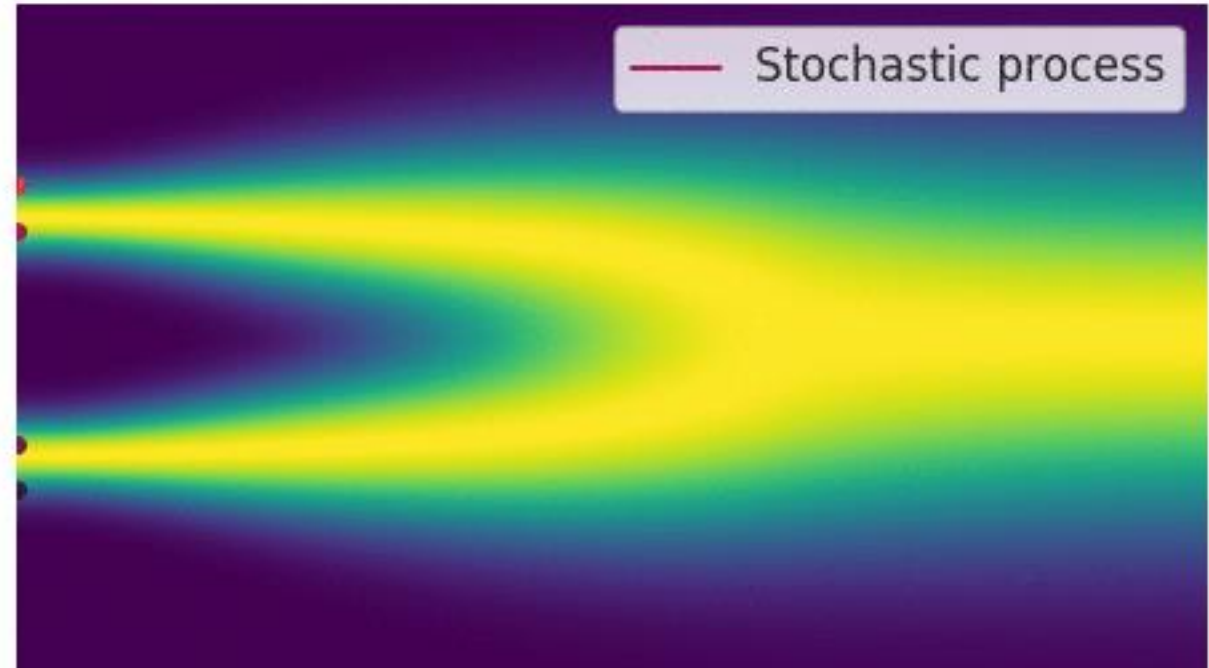
$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \boldsymbol{\mu}_t = \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \boldsymbol{\Sigma}_t = \beta_t \mathbf{I})$$

Константа

Обратный процесс

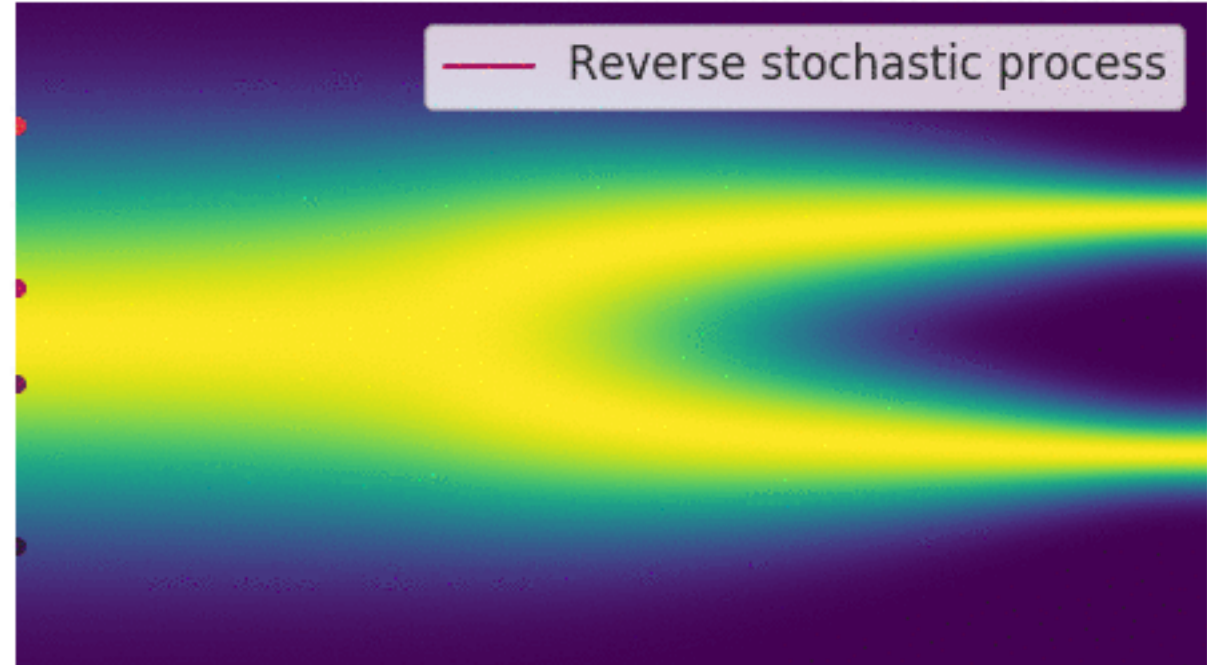


Демо: диффузия



Источник: <https://yang-song.net/blog/2021/score/>

Демо: обратный процесс

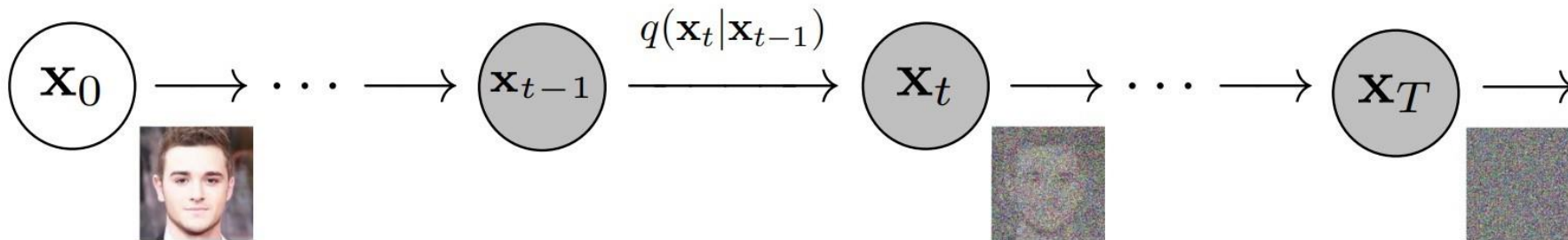


Источник: <https://yang-song.net/blog/2021/score/>

Диффузионные модели



Процесс диффузии (зашумления)

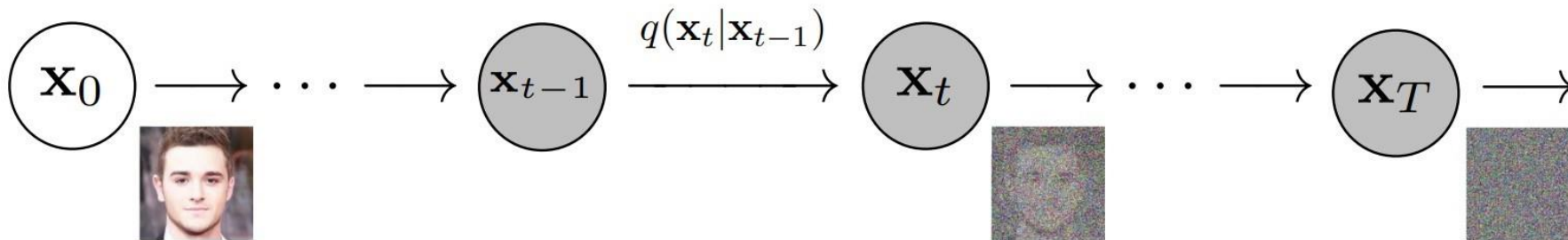


► $x_t = \sqrt{\alpha_t}x_{t-1} + \sqrt{1 - \alpha_t}\epsilon_t$

$$\epsilon_t \sim N(0, I), \quad \alpha_t = 1 - \beta_t$$

Мы сами так задали процесс

Процесс диффузии (зашумления)



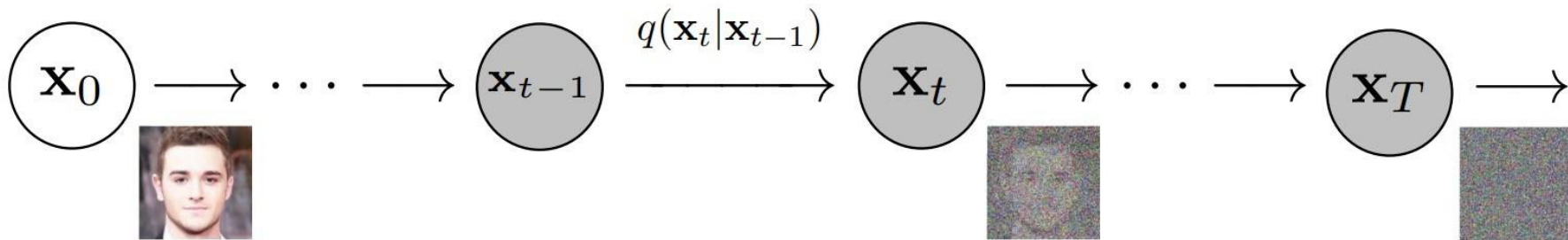
- ▶ $x_t = \sqrt{\alpha_t}x_{t-1} + \sqrt{1 - \alpha_t}\epsilon_t$

$$\epsilon_t \sim N(0, I), \quad \alpha_t = 1 - \beta_t$$

- ▶ $x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon$

$$\epsilon \sim N(0, I), \quad \bar{\alpha}_t = \prod_{i=1}^t \alpha_i$$

Процесс диффузии (зашумления)



$$x_t = \sqrt{\alpha_t} x_{t-1} + \sqrt{1 - \alpha_t} \epsilon_t$$

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$$

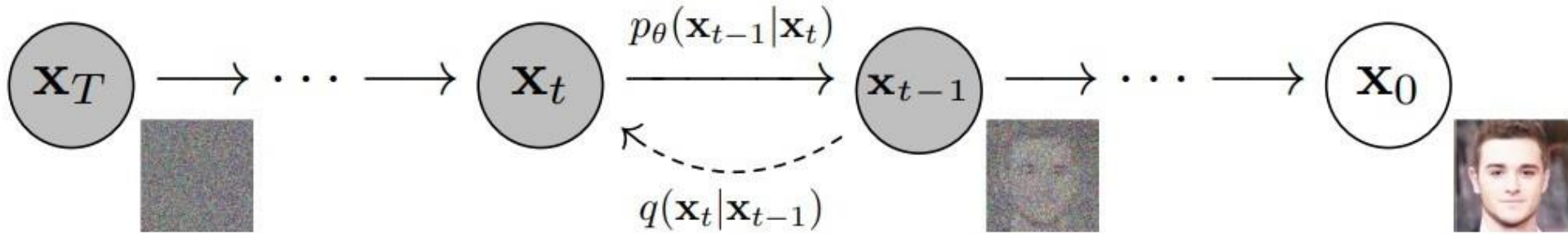
$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon \right) + \tilde{\beta}_t z$$

$$\epsilon_t \sim N(0, I), \quad \alpha_t = 1 - \beta_t$$

$$\epsilon \sim N(0, I), \quad \bar{\alpha}_t = \prod_{i=1}^t \alpha_i$$

$$z \sim N(0, I), \quad \tilde{\beta}_t = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t$$

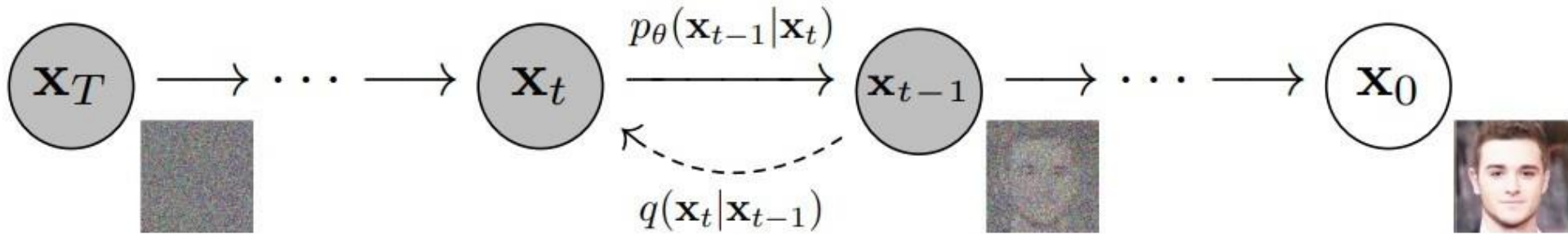
Обратный процесс



► $\hat{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(x_t, t) \right) + \sigma_t z$ $z \sim N(0, I), \quad \sigma_t = const$

Предсказываем нейронной сетью

Обратный процесс



- ▶ $\hat{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(x_t, t) \right) + \sigma_t z$ $z \sim N(0, I), \quad \sigma_t = const$

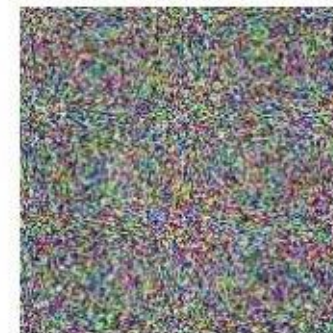
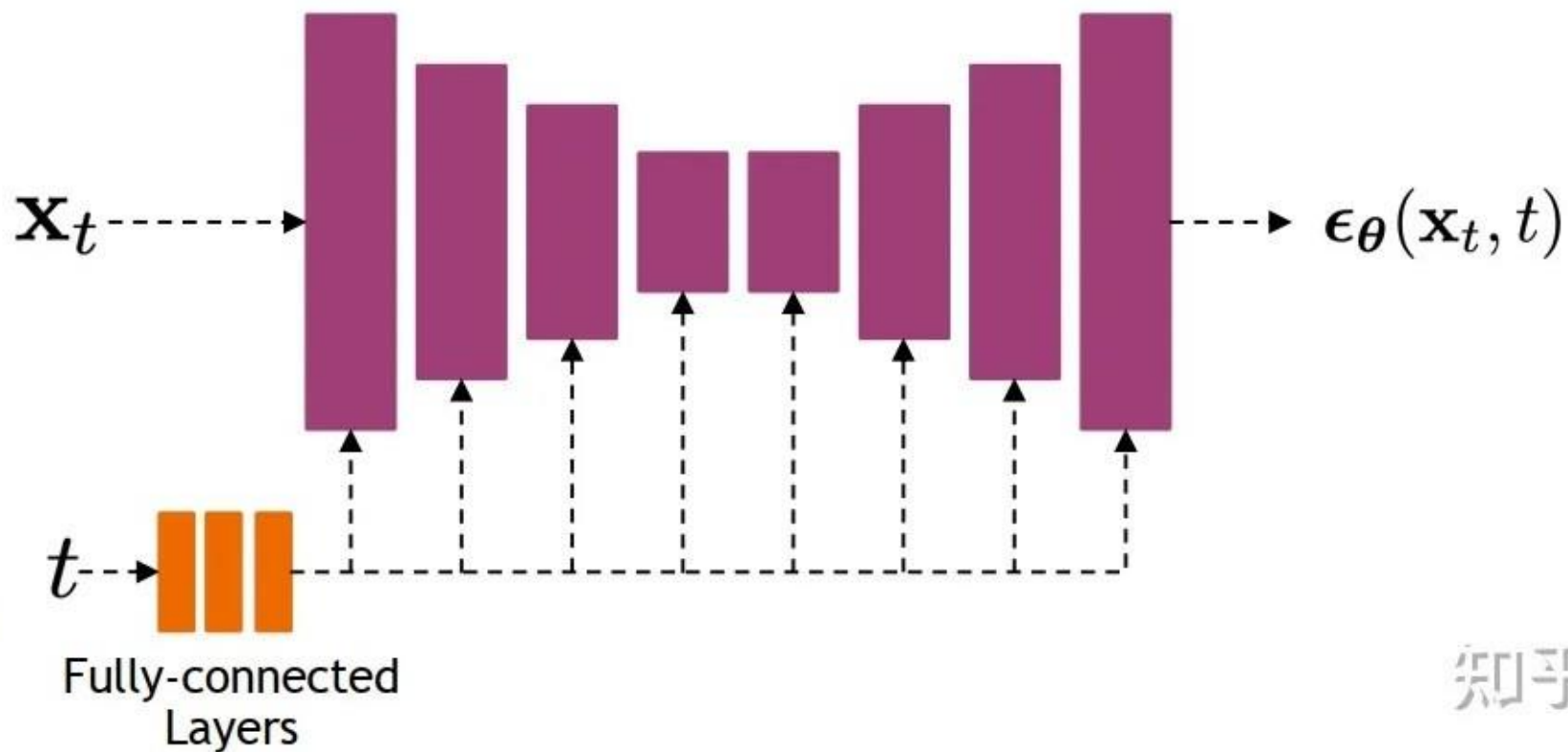
- ▶ Функция потерь для обучения:

$$L_t = \|x_{t-1} - \hat{x}_{t-1}\|_2^2 \propto \|\epsilon - \epsilon_\theta(x_t, t)\|_2^2 \rightarrow \min_{\theta}$$

Архитектура нейронной сети



Time Representation



知乎 @华年ss

Источник: <https://www.zhihu.com/question/536012286/answer/2683123893>

Алгоритм обучения

Algorithm 1 Training

- 1: **repeat**
 - 2: $\mathbf{x}_0 \sim q(\mathbf{x}_0)$
 - 3: $t \sim \text{Uniform}(\{1, \dots, T\})$
 - 4: $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
 - 5: Take gradient descent step on
$$\nabla_{\theta} \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_{\theta}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}, t) \right\|^2$$
 - 6: **until** converged
-

Алгоритм генерации

Algorithm 2 Sampling

```
1:  $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
2: for  $t = T, \dots, 1$  do
3:    $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$  if  $t > 1$ , else  $\mathbf{z} = \mathbf{0}$ 
4:    $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left( \mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \boldsymbol{\epsilon}_{\theta}(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$ 
5: end for
6: return  $\mathbf{x}_0$ 
```

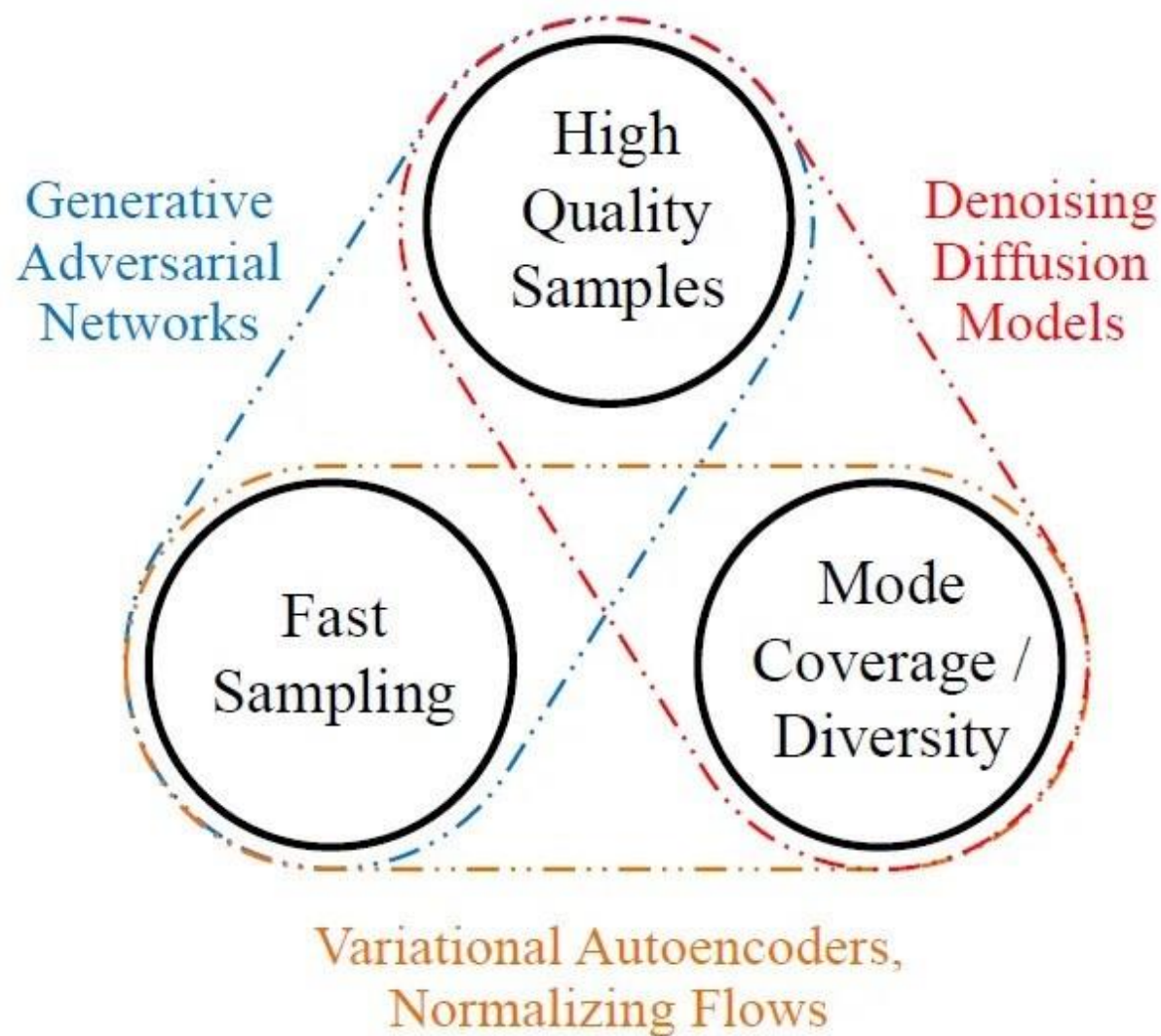
Значения гиперпараметров

- ▶ $T = 1000$
- ▶ $\beta_1 < \beta_2 < \dots < \beta_t < \dots < \beta_T$
- ▶ $\beta_1 = 0.0001, \beta_T = 0.02$
- ▶ $\sigma^2 = \beta_t$

Заключение



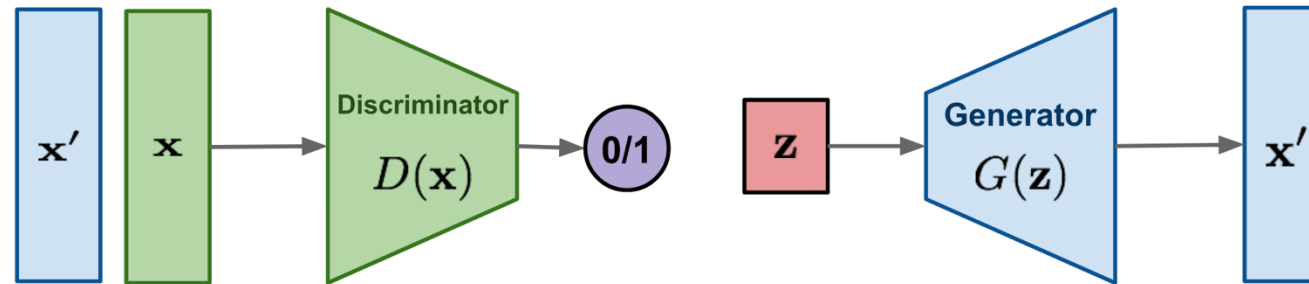
Generative learning trilemma



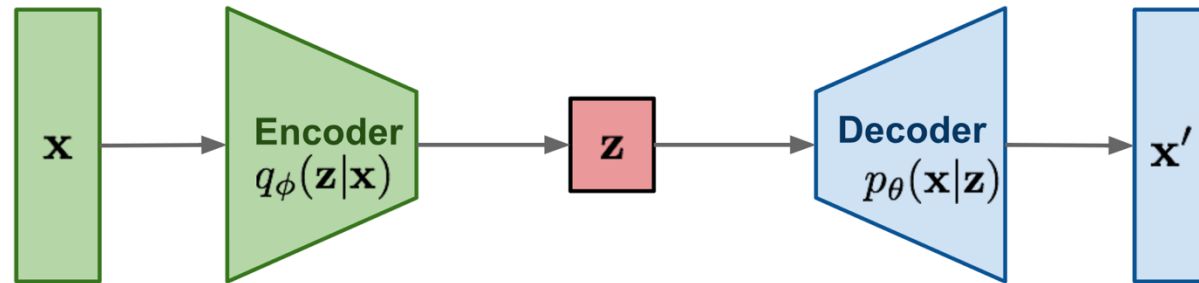
Source: <https://zhuanlan.zhihu.com/p/503932823>

Заключение

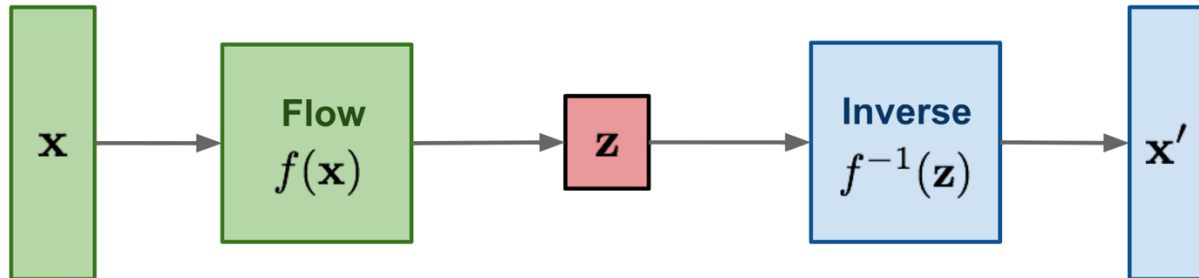
GAN: minimax the classification error loss.



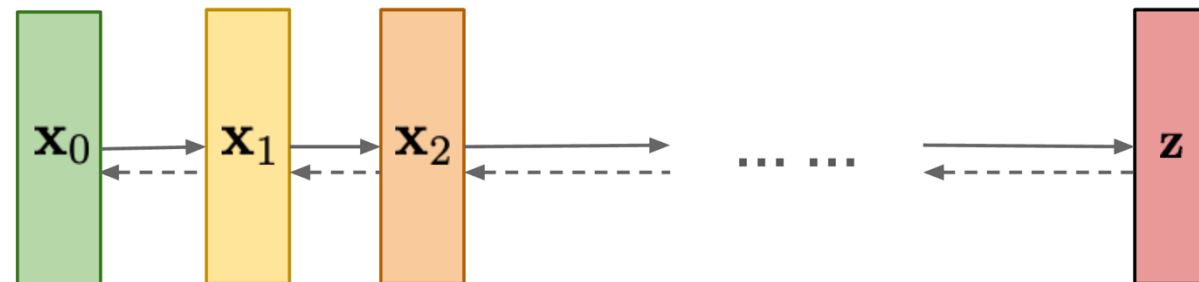
VAE: maximize ELBO.



Flow-based generative models:
minimize the negative log-likelihood



Diffusion models:
Gradually add Gaussian noise and then reverse



Практика

<https://colab.research.google.com/drive/1UG1zp0jqBIIKFwFOnYVEi9yqqldkjJb?usp=sharing>